

OSMOSIS ADVANCED PHY101 MCQS

PHY 101 – 100 LEVEL

**TOPICS: UNITS &
DIMENSIONS, SCALARS &
VECTORS,**

KINEMATICS, DYNAMICS, GRAVITATION.

1. The dimensional formula of impulse is:

- A. $[M L T^{-1}]$**
- B. $[M L^2 T^{-1}]$**
- C. $[M L T^{-2}]$**
- D. $[M L^2 T^{-2}]$**

2. Two vectors $A = 2i + 3j + k$ and $B = i - j + 4k$. The angle between them is:

A. $\cos^{-1}(5/\sqrt{42})$

B. $\cos^{-1}(6/\sqrt{42})$

C. $\cos^{-1}(7/\sqrt{42})$

D. $\cos^{-1}(8/\sqrt{42})$

3. A particle moves along x-axis: $x = t^3 - 6t^2 + 9t$. Its acceleration at $t = 2$ s is:

A. 0 m/s^2

B. 6 m/s^2

C. 12 m/s^2

D. 24 m/s^2

4. A force $F = 10 \text{ N}$ acts on a body of mass 2 kg . Its acceleration is:

A. 5 m/s^2

B. 2 m/s^2

C. 10 m/s^2

D. 20 m/s^2

5. The resultant of two equal vectors at 60° is:

- A. Equal to one vector**
- B. $\sqrt{3}$ times one vector**
- C. 2 times one vector**
- D. 0**

6. A body is dropped from height 45 m. Take $g = 9.8 \text{ m/s}^2$. Time to reach the ground is:

- A. 3 s**
- B. 4 s**
- C. 2 s**
- D. 5 s**

7. The dimensional formula of angular momentum is:

A. $[M L^2 T^{-1}]$

B. $[M L^2 T^{-2}]$

C. $[M L T^{-1}]$

D. $[M L T^{-2}]$

8. If the magnitude of vectors $A = 3i + 4j$ and $B = 4i - 3j$, the magnitude of $A + B$ is:

A. 5

B. 7

C. 8

D. 1

9. A particle moves with velocity $\mathbf{v} = 5t^2 \mathbf{i} + 3t \mathbf{j}$. Its speed at $t = 2 \text{ s}$ is:

A. $\sqrt{100 + 36} \text{ m/s}$

B. $\sqrt{25 + 9} \text{ m/s}$

C. $\sqrt{20 + 6} \text{ m/s}$

D. $\sqrt{10 + 3} \text{ m/s}$

10. A projectile is fired at 30° with speed 20 m/s. Maximum height is:

A. 5 m

B. 10 m

C. 15 m

D. 20 m

11. Two vectors $A = 6i + 2j$ and $B = -3i + 4j$. The scalar product $A \cdot B$ is:

A. 0

B. 6

C. 12

D. -10

12. The gravitational force between two masses m_1 and m_2 separated by distance r is F . If distance is doubled, new force is:

A. $F/2$

B. $F/4$

C. $2F$

D. $4F$

13. A particle moves along $y = x^2$. Slope of tangent at $x = 3$ is:

A. 3

B. 6

C. 9

D. 12

14. The speed of a satellite in circular orbit of radius r around mass M is:

A. $\sqrt{GM/r}$

B. $\sqrt{GM/r^2}$

C. $\sqrt{GMr}$

D. $\sqrt{GM/r}$

15. A particle moves with constant acceleration along x-axis. Displacement is proportional to:

A. t

B. t^2

C. t^3

D. $\sqrt{t}$

16. Two vectors $A = 5i + 2j$, $B = 3i + 6j$. The angle between them is:

A. 30°

B. 45°

C. 60°

D. 90°

17. The dimensional formula of pressure is:

A. $[M L^{-1} T^{-2}]$

B. $[M L T^{-2}]$

C. $[M L^2 T^{-2}]$

D. $[M L^{-2} T^{-1}]$

18. A particle moves along x-axis: $x = t^2 - 4t + 3$. Velocity at $t = 3$ s is:

A. 2 m/s

B. 0 m/s

C. -2 m/s

D. 4 m/s

19. A 5 kg body moves under force of 10 N. Distance moved in direction of force in 2 s is:

A. 4 m

B. 8 m

C. 10 m

D. 6 m

**20. A body slides down
frictionless incline of 30° .**

Acceleration along incline is:

A. g

B. $g \sin 30^\circ$

C. $g \cos 30^\circ$

D. $g/2$

**21. Two vectors $A = 3i + j + k$,
 $B = i + 2j - k$. Magnitude of $A \times B$ is:**

A. $\sqrt{14}$

B. $\sqrt{10}$

C. $\sqrt{12}$

D. $\sqrt{16}$

**22. The orbital period T of a
satellite of radius r is
proportional to:**

A. r

B. r^2

C. $r^{(3/2)}$

D. $\sqrt{r}$

23. The displacement of a particle moving with acceleration $a = 6t$ is:

A. $3t^2$

B. t^3

C. $6t^2$

D. $2t^3$

24. Two equal vectors at 120° produce a resultant.

Magnitude is:

A. Equal to vector

B. $\sqrt{3} \times \text{vector}$

C. $2 \times \text{vector}$

D. 0

25. A particle moves along $r(t) = t \mathbf{i} + t^2 \mathbf{j}$. Acceleration at $t = 2$ s is:

A. $2\mathbf{j}$

B. $4\mathbf{j}$

C. $2\mathbf{i} + 4\mathbf{j}$

D. $4\mathbf{i} + 2\mathbf{j}$

26. A body of mass 2 kg accelerates at 3 m/s^2 . Force applied is:

A. 2 N

B. 3 N

C. 6 N

D. 5 N

27. A projectile is fired horizontally from height h . Time to reach ground is:

A. $\sqrt{2h/g}$

B. $\sqrt{h/g}$

C. h/g

D. $2h/g$

28. The resultant of vectors $4\mathbf{i} + 3\mathbf{j}$ and $\mathbf{i} + 2\mathbf{j}$ is:

A. $5\mathbf{i} + 5\mathbf{j}$

B. $3\mathbf{i} + 1\mathbf{j}$

C. $5\mathbf{i} + 4\mathbf{j}$

D. $4\mathbf{i} + 5\mathbf{j}$

29. Two masses $m_1 = 4 \text{ kg}$, $m_2 = 6 \text{ kg}$ separated by 2 m .

Gravitational force ($G = 6.67 \times 10^{-11}$) is:

A. $4 \times 10^{-10} \text{ N}$

B. $8 \times 10^{-11} \text{ N}$

C. $6 \times 10^{-11} \text{ N}$

D. $2 \times 10^{-10} \text{ N}$

30. A particle moves with velocity $\mathbf{v} = 5t^2 \mathbf{i} + 3t \mathbf{j}$. Its acceleration vector is:

A. $10t \mathbf{i} + 3 \mathbf{j}$

B. $5t \mathbf{i} + 3 \mathbf{j}$

C. $10 \mathbf{i} + 6 \mathbf{j}$

D. $10t \mathbf{i} + 6 \mathbf{j}$

31. A body moves in circle radius $r = 2 \text{ m}$ at speed $v = 4 \text{ m/s}$. Centripetal acceleration is:

A. 4 m/s^2

B. 8 m/s^2

C. 16 m/s^2

D. 2 m/s^2

32. The angle between two perpendicular vectors is:

A. 0°

B. 45°

C. 90°

D. 180°

33. A particle moves under acceleration $a = t^2 + 2t$.

Velocity at $t = 2$ s (initially at rest) is:

A. 6 m/s

B. 8 m/s

C. 10 m/s

D. 12 m/s

34. Two equal forces act at point at 90° . Resultant magnitude is:

A. Equal to one force

B. $\sqrt{2} \times \text{force}$

C. $2 \times \text{force}$

D. $\frac{1}{2} \times \text{force}$

35. Acceleration due to gravity at height h above Earth's surface is:

A. $g(1 - h/R)$

B. $g(1 + h/R)$

C. $g(1 - h^2/R^2)$

D. $g(1 + h^2/R^2)$

36. The work done by force $F = 10 \text{ N}$ at 30° to displacement of 5 m is:

A. 25 J

B. 50 J

C. 10 J

D. 5 J

37. A particle moves along x-axis: $x = 3t^2 - t^3$. Velocity at $t = 2$ s:

A. 2 m/s

B. 4 m/s

C. 6 m/s

D. 8 m/s

38. Two masses connected over frictionless pulley, $m_1 = 5 \text{ kg}$ on incline 30° , $m_2 = 8 \text{ kg}$ vertical. Acceleration of system:

A. 1 m/s^2

B. 2 m/s^2

C. 3 m/s^2

D. 4 m/s^2

39. A body of mass 1 kg moving at 10 m/s hits a wall

and rebounds at 6 m/s.

Impulse is:

A. 4 N·s

B. 16 N·s

C. 8 N·s

D. 10 N·s

40. Two vectors A and B, $|A| = 5$, $|B| = 12$, angle = 90° .

Magnitude of $A + B$:

A. 13

B. 17

C. 7

D. 10

41. A particle moves under $F = ma + bv$, b constant. Velocity vector as function of time:

A. Exponential decay

B. Linear

C. Quadratic

D. Sinusoidal

42. The resultant of vectors $3\mathbf{i} + 4\mathbf{j}$ and $5\mathbf{i} + 12\mathbf{j}$ is:

A. $8\mathbf{i} + 16\mathbf{j}$

B. $8\mathbf{i} + 12\mathbf{j}$

C. $8\mathbf{i} + 14\mathbf{j}$

D. $8\mathbf{i} + 10\mathbf{j}$

43. A particle moves along $y = x^3$. Tangent perpendicular to velocity at:

A. $x = 0$

B. $x = 1$

C. $x = 2$

D. $x = 3$

44. Escape velocity from planet radius R , g surface gravity:

A. $\sqrt{2gR}$

B. $\sqrt{gR}$

C. gR

D. $2gR$

45. Two satellites orbit same planet at R_1 and R_2 , same mass. If orbital period $T_1 = T_2/2$, ratio R_1/R_2 is:

A. $1/\sqrt{2}$

B. $1/4$

C. $1/2$

D. $1/\sqrt{8}$

46. Velocity of a particle moving along x-axis: $v = 3t^2 - 4t + 2$. Time when velocity is zero:

A. 1 s

B. 2 s

C. 3 s

D. 4 s

47. Two forces act at 120° on a particle. Resultant $R = 20\text{ N}$. If one force is 12 N , the other is:

A. 16 N

B. 18 N

C. 20 N

D. 15 N

48. A particle slides down frictionless hemisphere. Leaves surface at:

A. $h = \frac{2}{3} R$

B. $h = R/2$

C. $h = R$

D. $h = 1/3 R$

49. Magnitude of cross product of $A = 2i + 3j$ and $B = 5i - j$ is:

A. 13

B. 10

C. 11

D. 12

50. A particle moves with acceleration $a = 6t$. Initial velocity zero. Displacement after $t = 3$ s is:

A. 27 m

B. 54 m

C. 36 m

D. 18 m

BEST OF LUCK!!!